The Constrained Future

A Testable Information-Constraint Principle for Persistent Organization

Douglas Clark

Independent Researcher

Corresponding author: clarkdd@gmail.com

Article type: Hypothesis Paper

Abstract

Living organization persists while its material components are replaced. We formulate the Information-Constraint Principle (ICP) as an operational hypothesis about such persistence: physically effective information is an interaction-formed, embodied constraint on an explicitly indexed family of reachable or probable future responses, and it persists only when its causal role is re-instantiated in successor carriers. A critical comparison with viability theory, the adjacent possible, empowerment, counterfactual semantic information, computational mechanics, constructor theory, and feedback thermodynamics shows that most future-option language is established prior art. ICP's remaining content is a persistent-carrier test requiring formation, sufficiency, carrier intervention, succession across replacement, and representation robustness. We formalize the adjacent possible as the novelty-bearing projection of an indexed reachable set and show that a signed support deformation equals forward Kullback–Leibler divergence under pruning and negative reverse divergence under enabling. Existing models of the adjacent possible postulate the step that opens new possibilities; ICP makes that step an intervention on an embodied carrier and predicts, unlike triggering models, that opened possibilities close when their carrier is removed, as permissive substitutions in glucocorticoid-receptor evolution illustrate. Applied to yeast *INO1* transcriptional memory, the test shows that a chromatin record survives several rounds of replication-coupled carrier replacement while depending on a maintenance system, but that succession of the functional role has not been tested; we specify the experiment that would test it and the nucleosome-modification model it must be compared against. ICP is a falsifiable organizing hypothesis, not an established law: its decisive test is whether carrier interventions predict held-out changes in future responses beyond complete native dynamics.

Keywords: information constraint; adjacent possible; constraint closure; epigenetic memory; carrier succession; persistent organization

1 The persistence problem

Living organization persists while its material components are replaced. Enzymes are degraded and resynthesized, nucleosomes are diluted at every replication fork, and cells in a tissue turn over

while the tissue continues to perform its role. The motivating question is how the consequence of an interaction can persist after the object or arrangement that first carried it has changed or disappeared, without slipping between incompatible meanings of "information." None of these examples requires information to float free of physical realization. They distinguish persistence beyond an *original* carrier from existence without *any* carrier.

Embodied-information accounts in theoretical biology already treat patterns as constraints on physical processes and emphasize their historical accumulation [12]. Constraint-closure theory models conserved structures that act on processes while depending on one another for their maintenance [27], and the theory of the adjacent possible treats organization as enabling futures that were previously inaccessible [18, 19]. A new proposal must therefore earn its place through operational criteria, not through the language of embodiment or constraint alone.

We state the proposal as follows:

> **Information-Constraint Principle (ICP).** Physically effective information is interaction-formed, embodied constraint on an explicitly indexed family of reachable or probable futures; it persists when that constraint role is lawfully re-instantiated beyond its original carrier.

ICP arose within a broader information-first modelling program whose artificial-life testbed, IOTA, is reported separately [5]. The present paper does not depend on that program's physical ontology. It asks a narrower question: does the combination of historical formation, physical embodiment, sufficiency for a declared family of future responses, causal carrier manipulation, succession across replacement of the original carrier, and robustness to representation yield a testable claim that existing theories do not already make?

The phrase "constrains the future" is easy to satisfy. Given a complete physical state and law, virtually every state difference can alter a future trajectory. Control theory studies reachable sets; computational mechanics conditions futures on pasts; and semantic-information work erases correlations and measures effects on later viability. If ICP merely renames these objects it may be useful language, but it is not a new principle. The paper is accordingly adversarial: it formalizes ICP (Section 2), tests it against its closest native formalisms (Section 3), develops its relation to the adjacent possible (Section 4), applies it to a well-characterized biological memory (Sections 5–6), and grades what survives (Sections 8–9).

2 Formalism

This article is a critical integrative synthesis and operational proposal, not a systematic review. A project registry records query strata, primary-source checks, source notes, competing explanations, falsification criteria, and a claim–evidence matrix; searches were expanded adversarially whenever a field offered a likely redundancy account. Claims are labelled as established field results, project evidence, synthesis inferences, or ICP hypotheses, and project papers are never counted as independent replication.

2.1 Indexed future-law families

For a candidate retained state or record m, intervention family Π, observable algebra $\mathcal{A}$, boundary and resolution declaration B, resource declaration C, and horizon τ, define

$$\mathcal{E}_\tau(m; \Pi, \mathcal{A}, B, C) = \left\{ P^\pi_{\mathcal{A}}(F_{t:t+\tau} \mid m, B, C) : \pi \in \Pi \right\}. \tag{1}$$

The corresponding reachable support for policy π is

$$\mathcal{R}^\pi_\tau(m) = \operatorname{supp} P^\pi_{\mathcal{A}}(F_{t:t+\tau} \mid m, B, C), \tag{2}$$

with topology and tolerance specified for continuous spaces.

Equation (1) is not itself an ICP law; it is shared comparison language. A *constraint* requires a matched contrast between families, for example a history-formed carrier m and a reference m_0 prepared with the same declared ordinary state variables and resources:

- a **hard constraint** changes reachable support on a common path or outcome space;
- a **soft constraint** changes weights while support may remain fixed; and
- an **operational constraint** changes a reachability or convertibility preorder under the same allowed operations.

2.2 A relative-entropy constraint diagnostic

For a declared comparison m versus m_0 at x, one distribution-sensitive member of the future-family comparison is

$$\mathfrak{C}^\pi_{x,\tau}(m \| m_0; \mathcal{A}, B, C) := D_{\mathrm{KL}}(P^\pi_{\mathcal{A}}(F_{t:t+\tau} \mid m, B, C) \| P^\pi_{\mathcal{A}}(F_{t:t+\tau} \mid m_0, B, C)), \tag{3}$$

provided both laws use the same outcome space, intervention, horizon, measure, and coarse-graining, with the first law absolutely continuous with respect to the second [22]. This is an indexed diagnostic, not the definition of ICP: the full family retains outcome identity, topology, interventions, and operational order that a single scalar discards. For a soft constraint it is the natural quantity; for a hard constraint it has an exact support specialization. If the post-intervention law is uniform on $\mathcal{R}^{\mathrm{post}}_x \subseteq \mathcal{R}^{\mathrm{prior}}_x$ and the reference law is uniform on $\mathcal{R}^{\mathrm{prior}}_x$, then

$$D_{\mathrm{KL}}(U^{\mathrm{post}}_x \| U^{\mathrm{prior}}_x) = \log \frac{|\mathcal{R}^{\mathrm{prior}}_x|}{|\mathcal{R}^{\mathrm{post}}_x|}. \tag{4}$$

The direction matters: the reversed divergence is infinite under strict support pruning.

2.3 The adjacent frontier and signed possibility-space flux

The adjacent possible becomes operational only after its outcome types and notion of "one step" are declared. Let $r_\ell : \Omega \to \mathcal{K}_\ell$ be a frozen scale-ℓ map from micro-outcomes to registered types or roles, and let $\mathcal{V}_t^\ell$ be the types actualized by time t. Define

$$\mathcal{J}_{t,\tau}^{\pi,\ell}(m_t) := r_\ell(\mathcal{R}_\tau^\pi(m_t)) \setminus \mathcal{V}_t^\ell. \tag{5}$$

When τ is one registered transition, this is the novelty-bearing sector of Kauffman's adjacent possible: types not yet actual but reachable in one step from the actual [18, p. 142]. It depends on r_ℓ, π, τ, B, and C; changing a classifier, resource budget, or policy can change the set.

For matched pre/post supports R_0 and R_1, retain the identity of the change rather than only its cardinality:

$$K = R_0 \cap R_1, \qquad E = R_1 \setminus R_0, \qquad L = R_0 \setminus R_1, \tag{6}$$

the retained, enabled, and lost possibilities. A signed log-volume deformation is $\chi(R_1 : R_0) := \log[\mu(R_0)/\mu(R_1)]$. For nonempty finite uniform nested supports it gives the exact dual pair

$$\chi(R_1 : R_0) = \begin{cases} D_{\mathrm{KL}}(U_{R_1} \| U_{R_0}), & R_1 \subseteq R_0 \quad \text{(pruning)}, \\ -D_{\mathrm{KL}}(U_{R_0} \| U_{R_1}), & R_0 \subseteq R_1 \quad \text{(enabling)}. \end{cases} \tag{7}$$

Thus $\chi > 0$ denotes net contraction and $\chi < 0$ net expansion. For non-nested supports χ is only a net-volume contrast and can vanish under complete identity turnover, so the ledger in Eq. (6) must be reported alongside it.

This duality makes one biologically familiar pattern precise. A retained carrier may restrict immediate microstate options while enabling more long-horizon role types,

$$\chi_{\mathrm{micro},1} > 0, \qquad \chi_{\mathrm{role},h} < 0 \quad (h \geq h_\star), \tag{8}$$

under the same randomized carrier intervention. Enabling-constraint and scaffolding accounts already anticipate this qualitative pattern [14, 19]; it becomes an ICP result only if $h_\star$ is predicted from independently measured carrier lifetimes and confirmed on held-out systems.

2.4 The persistent-carrier criterion

A proposed informational carrier M passes the ICP criterion only if four gates are met:

Formation. Controlled histories create the candidate distinction.

Sufficiency. The distinction is sufficient for the declared future-law family to a frozen error tolerance.

Carrier intervention. Ablating, replacing, transplanting, or scrambling the proposed carrier

changes a preregistered component of the future family, while matched shams and resource controls do not.

Robustness. The result survives adequate state augmentation and reasonable representation, partition, and coarse-graining changes, or its representation dependence is stated as a boundary.

Persistence across carrier succession adds one further demand. Let $c_0, c_1, \ldots$ be physical realizations at successive times. They instantiate the same role only relative to a declared equivalence relation $\sim_{\Pi,\mathcal{A},B,C,\tau}$ under which replacement preserves the relevant future-response family. Material disjointness alone does not establish sameness of role; sameness of an analyst's label does not establish physical transport.

2.5 Novelty and failure tests

The strongest null is complete-state ordinary dynamics: once every causally relevant variable, law, input, boundary, and control is included, there is no additional historical variable. ICP earns distinct content only by delivering at least one of the following beyond the best native model: a minimal sufficient physically embodied equivalence class; a representation-stable carrier-succession invariant; a new impossibility or conversion result; a demonstrably superior compression of intervention–response families; or a novel preregistered prediction. Disappearance under adequate state augmentation, failure of matched carrier interventions, or dependence on one convenient representation would lower or defeat the relevant claim.

3 Closest native formalisms and the redundancy test

The literature that matters most for ICP is the literature capable of making it redundant.

3.1 Reachability and viability

Hamilton–Jacobi reachability characterizes finite-horizon backward-reachable sets of nonlinear differential games as sublevel sets of viscosity solutions [25], and viability theory identifies states from which some admissible evolution remains inside a declared constraint set [30]. These are mature theories of accessible and survivable futures. A reachability calculation is, however, given the state, law, controls, and horizon; it need not ask how a constraint role was formed, where it is physically realized, or whether it survives replacement of its first carrier. That upstream question is the only place ICP can add content.

3.2 The adjacent possible

The adjacent possible is the closest native formalism and is treated separately in Section 4.

3.3 Empowerment and semantic information

Empowerment is the channel capacity from action sequences to later sensed states [20], and causal entropic forcing builds a path-entropy objective into an effective force [41]. These defeat any claim that ICP discovered the valuing of future options. ICP does not assert that systems maximize optionality: a constraint may reduce, redirect, or reorganize futures, and scalar option values collapse outcome identity.

Kolchinsky and Wolpert provide the closest formal rival. They define semantic information as system–environment correlation whose controlled scrambling lowers a declared future viability [21]; the intervention, boundary, horizon, and viability function are constitutive. Causal-information measures likewise require post-intervention distributions [1], and predictive association alone is not a carrier test [17]. ICP is broader only in that its target may be any registered future-response role rather than viability, and in its emphasis on formation and succession. On a self-maintenance task an ICP analysis should specialize to semantic information, not rename it.

3.4 Computational mechanics and constructor theory

Causal states are equivalence classes of pasts with the same conditional distribution over futures and give a minimal optimal predictive representation [31]; epsilon-transducers extend this to input-dependent futures [2]. This is almost exactly the $H \to M \to P(F \mid M)$ part of ICP. The difference is physical: an inferred causal state need not be localized, manipulable, or interventionally defined, whereas the carrier criterion asks whether an embodied realization is necessary and sufficient and whether its replacement preserves the role. Constructor theory supplies a law-level language of possible and impossible tasks, including information as copyable and distinguishable variables [7, 8], but not a history-conditioned process.

3.5 Feedback thermodynamics

Feedback thermodynamics shows that a retained measurement changes later work bounds [29], demonstrated experimentally with a Brownian particle [35] and made fully autonomous in a bit-tape demon [24]. It supports causal utility and closed accounting of memory, but energy, baths, and protocols enter as supplied structure.

3.6 Result of the audit

The generic proposition that present organization constrains the future is preempted. The viable target is narrower:

> An interaction forms a physically embodied constraint role; the original carrier ceases to instantiate it; a successor realization preserves the registered intervention–response role; and perturbing that successor changes the future family under matched controls.

Biology is the natural place to test this statement, because component turnover with persisting function is ubiquitous there and carrier interventions are experimentally routine.

4 The adjacent possible and its carriers

Of the native formalisms, the adjacent possible is the closest to ICP and the one in which ICP has the clearest opportunity to contribute. This section sets out what the formal theory already establishes, identifies the step it leaves unspecified, and states a prediction that separates a carrier-based account from the existing models.

4.1 What is formalized

Kauffman defined the chemical adjacent possible as the molecular species absent from the actual but synthesizable from it in one reaction step [18, p. 142], and later argued in this journal that enabling constraints open unprestatable possibilities in biological evolution [19, 28]. Tria, Loreto, Servedio, and Strogatz gave the idea its first minimal mathematical form [36]. In their urn model with triggering, an element drawn from an urn is returned together with ρ copies (reinforcement); if and only if the element is new, $\nu + 1$ brand-new elements are also added, so that $\nu + 1$ is the size of the adjacent possible opened by that novelty. The balance of the two processes fixes the statistics: the number of distinct elements grows as $D(N) \sim N^{\nu/\rho}$ for $\nu < \rho$, as $N \log N$ for $\nu = \rho$, and linearly for $\nu > \rho$, with Zipf's law $f(R) \sim R^{-\rho/\nu}$. A semantic-triggering variant reproduces the clustering of related novelties observed in Wikipedia edits, music listening, annotation tags, and texts [36].

The framework has since been extended in several directions. Explorations of the adjacent possible in social contact space reproduce the rate at which individuals acquire new acquaintances and the degree, clustering, and community structure of real networks [37]. Waves of novelties in which newcomers are not suppressed by established elements have been quantified across several systems [26]. A time-dependent urn with triggering, with reinforcement and triggering parameters that vary in time, reproduces the logarithmic novelty growth observed in gene mutations and patent codes and has a critical region in which Heaps' and Zipf's laws coexist [4]. Edge-reinforced walks place discovery on an explicit concept network [16], and higher-order novelty counts first occurrences of combinations, distinguishing sequences with similar first-order discovery rates [9]. Combinatorial TAP models generate long plateaus followed by explosive growth [32], while long-run product data show resource-constrained rather than unrestricted combinatorial discovery [34]. Fariña and Vallejo give a geometric formulation, a biological cosmological constant proportional to the squared fractional expansion rate of Fisher-information volume [11].

ICP therefore cannot claim history-dependent accessibility, triggered support growth, correlated novelty, a plateau-to-explosion transition, the Heaps and Zipf regularities, or a geometric measure of expansion. All of these are prior art.

4.2 What is left unspecified

In each of these models the opening step is postulated. The triggering parameter ν, or its time-dependent generalization, is fitted to data, and the elements it adds are featureless new colours or nodes; what in the system produces them lies outside the model. Two further features follow from this construction. First, the basic urn never removes elements, so possibilities once opened stay open indefinitely: in the ledger of Eq. (6) only the enabled set E is populated, and the lost set L is empty by construction. Second, the opened possibilities depend on the history only through the count and order of past novelties, not through any structure that remains physically present.

Kauffman's enabling constraints name the missing element without making it operational. This is the step ICP's carrier criterion is designed for. In its terms, an account of how the adjacent possible expands must identify an embodied carrier m, formed by the novelty event, whose presence makes the frontier $\mathcal{J}^{\pi,\ell}_{t,\tau}$ of Eq. (5) larger than it is under a matched reference m_0. The triggering size then becomes an intervention–response quantity,

$$\nu_{\mathrm{C}}(m) := \left|\mathcal{J}^{\pi,\ell}_{t,\tau}(m)\right| - \left|\mathcal{J}^{\pi,\ell}_{t,\tau}(m_0)\right|, \tag{9}$$

measured by ablating, restoring, or replacing the carrier rather than fitted to a sequence, and reported together with the identities of the enabled and lost types. ICP does not supply the mechanism that makes ν_{C} positive in a given system; that is chemistry, biophysics, or sociology. It supplies the test that would identify the mechanism, and the accounting under which the identification can fail.

4.3 A discriminating prediction

The two accounts differ on one point that can be tested. In an urn model, possibilities opened by a novelty persist whatever happens afterwards. On the carrier account, an opened possibility persists only while its carrier, or a role-equivalent successor, remains realized:

> **Carrier dependence of the adjacent possible.** If the carrier formed by a novelty event is removed after the event, without a successor, the possibilities it opened leave the frontier, and they return if the carrier is restored.

A pure triggering model predicts no change in the frontier under this intervention, because the history of novelties is unchanged. A complete mechanistic model of a particular system may of course predict the same as the carrier account; the prediction distinguishes ICP from the phenomenological models of the adjacent possible, not from complete-state dynamics.

Molecular evolution already provides a case in which the prediction holds. During the evolution of cortisol specificity in the ancestral glucocorticoid receptor, permissive substitutions were functionally neutral in isolation but had to be present for the large-effect function-switching mutations to be tolerated [15]. In the terms above, the permissive substitutions are the carrier, the set of tolerable function-switching mutations is the enabled set E, and the dependence of tolerance on their presence is the carrier dependence of that frontier. The same study found that permissive genotypes were

extremely rare: a library of thousands of variants of the ancestral protein recovered the historical permissive substitutions but no alternatives, because a permissive mutation must stabilize specific local structure, preserve the energetic balance between conformations, and be compatible with both ancestral and derived structures [15]. Contingency in which few paths are open also appears in the exhaustive reconstruction of TEM β-lactamase trajectories [40]. These are cases in which the opener of an adjacent possible has been identified physically and in which its ablation is experimentally meaningful.

The pattern also connects to the crossover of Eq. (8). Permissive substitutions restrict nothing measurable at the level of immediate receptor function, yet they enlarge the set of later functional changes; a carrier that prunes microstates while enlarging the role frontier is the general form of the same observation.

4.4 From statistics to carriers

The urn models explain the statistics of discovery from a small number of parameters. The carrier account asks what sets those parameters. If it is correct, the triggering and reinforcement rates of a system should be predictable from measured properties of its carriers: how often novelty events form a carrier, how large a frontier each carrier opens, and how long carriers survive or are succeeded. A time-dependent triggering rate [4] would then be read as a changing population of realized carriers. Making that link quantitative in one system, by predicting ν and its time dependence from carrier measurements before fitting the sequence, is the most direct way in which ICP could become an explanation of the adjacent possible rather than a redescription of it.

5 Biological records and their ordinary rivals

Biology already supplies the architecture

$$\text{history} \rightarrow \text{embodied record} \rightarrow \text{changed later response}, \tag{10}$$

including cases in which the initiating event or molecule is gone. In bacterial acquired immunity, phage exposure installs a sequence record in a CRISPR locus, and adding or deleting matching spacers changes later resistance [3]. Exhaustive construction of all 32 combinations of five TEM β-lactamase substitutions found only 18 of 120 direct mutational paths accessible under the measured landscape [40], and ancestral reconstruction of the glucocorticoid receptor showed that rare permissive substitutions made later functional changes tolerable [15]: historical contingency acting on operational accessibility, a hard and an operational constraint in the terms of Section 2. Constructive origin-of-life work supplies controls in both directions. Cooperative RNA networks can outgrow selfish cycles [38], while compositional autocatalytic networks show that self-maintenance need not yield selected heredity [39].

Constraint-closure theory gives these cases a common organizational reading: an enzyme may

constrain one reaction while being produced and degraded on another timescale, so constraints persist through the turnover of their realizers [27]. Whether closure grounds function or normativity remains contested [6, 13].

Every example nonetheless has a sufficient ordinary mechanism: sequence and base pairing, chromatin chemistry, protein structure, population dynamics, and reaction kinetics. A state description that omits the spacer, mark, or enzyme will appear history-dependent; an adequate physical state includes the carrier. Generic future-shaping is therefore not ICP novelty. What biology does provide is experimentally controllable systems in which the full criterion of Section 2.4—including succession—can be applied and, crucially, can fail. The next section applies it to one such system in detail.

6 Worked case: transcriptional memory through chromatin succession

6.1 The system

Budding yeast cells previously starved of inositol reactivate the *INO1* gene faster on a second starvation than cells without that history. The memory requires a cis-acting DNA element (the memory recruitment sequence, MRS), the transcription factor Sfl1 bound to it, interaction with the nuclear pore complex, incorporation of H2A.Z, and a memory-specific dimethylation of histone H3 lysine 4 (H3K4me2) [23, 33]. The mark is written by a form of COMPASS lacking the Spp1 subunit and requires SET3C, a complex that binds H3K4me2 and interacts physically with the writer; the authors propose this writer–reader coupling as the mechanism for re-incorporating the mark after DNA replication [33].

Sump and colleagues separated establishment from inheritance by inducible degradation of Sfl1. When Sfl1 was degraded after memory had been established, peripheral localization was lost but H3K4me2 remained at the promoter for roughly three to four generations; degrading Set3 together with Sfl1 caused rapid loss of the mark [33]. The authors conclude that memory duration is governed by the number of divisions or replication cycles, reflecting dilution, degradation, or imperfect fidelity of inheritance.

6.2 Mapping onto the criterion

Each replication fork distributes parental nucleosomes between the daughter chromatids and incorporates new ones, so the physical realizations $c_0, c_1, c_2, \ldots$ of the mark are progressively disjoint in material composition. This is the situation the succession clause of Section 2.4 was written for. Table 1 gives the mapping.

Table 1: The persistent-carrier criterion applied to *INO1* transcriptional memory. "Shown" refers to published results [23, 33]; "untested" marks components of the criterion that no published experiment yet addresses.

Element	Instantiation	Status
History m vs. m_0	Prior inositol starvation vs. never-starved cells in the same current medium	Shown
Future family $\mathcal{E}_\tau$	Distribution of *INO1* re-induction times after re-starvation, indexed by the delay $\pi \in \Pi$	Shown (faster reactivation)
Constraint type	Soft: naive cells also induce, so the support is unchanged and the weights shift; Eq. (3) applies directly	Inferred
Formation	Sfl1 and pore interaction required to establish the mark	Shown
Carrier intervention	Removing writer/reader components (COMPASS, SET3C) abolishes the mark and faster reactivation; matched shams required	Shown, shams partial
Succession	H3K4me2 re-incorporated onto new nucleosomes for 3–4 generations after initiator removal	Shown for the mark
Role equivalence $\sim$	Successor cells whose re-induction distribution lies within a declared tolerance of the memory distribution	Untested across succession
Role exhaustion	Mark lost after finitely many divisions while the lineage lives: $\mathcal{R}_C = \varnothing$ for the role, not the substrate	Shown for the mark

6.3 What the case shows and does not

The case passes formation, sufficiency, and carrier intervention for the mark, and it demonstrates succession of the *record* through several rounds of replication-coupled carrier replacement. It does not yet demonstrate succession of the *role*. Sfl1 is itself required to express the faster reactivation, so while Sfl1 is absent the future-response family cannot be assayed: the inherited mark is a latent record whose functional reading has not been measured after succession. This distinction between persistence of a record and persistence of a role is exactly the one ICP exists to enforce, and the published literature does not need ICP to draw it. It does, however, make the missing experiment precise.

Prospective test. Establish memory, degrade Sfl1, allow $k = 0, 1, \ldots, 5$ generations of latent inheritance, restore Sfl1, and measure re-induction after re-starvation. ICP's succession clause predicts that the divergence $\mathfrak{C}_k$ between memory and naive re-induction-time distributions decays with k at the per-generation rate of H3K4me2 retention at the locus, and that degrading Set3 only during the latent interval abolishes $\mathfrak{C}_k$ while a matched sham degradation does not. If restored Sfl1 re-establishes fast reactivation independently of the inherited mark, the role was not carried by the successor record and the case fails the succession gate.

The ordinary rival. Nucleosome-modification models with recruited positive feedback already predict that the stability of a chromatin memory is set by the balance between feedback-driven

re-modification and replication-coupled dilution, with bistable, effectively heritable states above a feedback threshold [10]. A complete chromatin model of this kind, with the writer and reader machinery included, fixes the future family in the sense of Eq. (11). We expect it to predict the outcome of the prospective test, and we do not claim that ICP explains *INO1* memory better than chromatin dynamics does.

The case is therefore a positive control for the criterion rather than evidence for a new mechanism. A criterion that failed to classify a known, mechanistically understood memory correctly would be disqualified; one that classifies it correctly, separates record from role, and specifies the untested experiment has earned the right to be applied where the mechanism is not known. The same table can be filled in for CRISPR spacers, prion conformational templates, and artificial systems, which makes them comparable as instances of one question.

7 An artificial-life testbed

The companion study applies the same criterion to a stochastic finite local lattice model used as an artificial-life testbed [5]. Its central result concerns a relational connectivity role rather than a marked payload: in three prospectively replayed events across two seeds, tracked regions lost connectivity and regained it through a causally necessary successor bridge sharing no cells with its predecessor, and removing the successor destroyed the restored relation. Event-resolved vetoes further showed that alternative successors are frequently available but that an available future is causally relevant without being sufficient. Within that model Eq. (4) holds exactly for its support measure. These results establish possibility within one engineered law, not use by living systems; they illustrate hard constraints, complementing the soft constraint of Section 6.

8 Adversarial audit and verdict

8.1 The complete-state challenge

Let Z_t be a complete physical specification at time t, including all degrees of freedom relevant at the chosen precision, their correlations, boundary conditions, resources, controls, and dynamics. Then

$$\mathcal{E}_\tau = \Phi_\tau(Z_t; \Pi, \mathcal{A}, B, C). \tag{11}$$

On this account a memory, chromatin mark, or regulatory state matters only because it is part of Z_t; enlarging the state until it is sufficient removes any independent history variable. This null is not defeated by showing that two visibly similar systems behave differently, by finding mutual information between past and future, or by a compact description alone. ICP becomes distinct only if its carrier-equivalence construction does useful work beyond Eq. (11): a minimal manipulable sufficient class, an invariant across genuine succession, a validated compression of intervention–response laws, or a preregistered prediction missed by the best complete-state model.

The *INO1* case shows the criterion operating correctly; it does not supply such a result.

8.2 Failure modes that remain live

1. **State-augmentation collapse.** A claimed historical effect disappears when the carrier or environment is adequately represented, leaving ICP as vocabulary for ordinary memory.

2. **Representation dependence.** The carrier exists only under a convenient partition, encoding, or coarse-graining.

3. **Inferential–physical equivocation.** An observer's loss of uncertainty is mistaken for a change in the system's possibilities.

4. **Generic-dynamics redescription.** Without the formation, intervention, succession, and robustness gates, ICP collapses into reachability, viability, or semantic-information theory.

5. **Model artifact.** Effects in an artificial system depend on one update rule, scheduler, or topology.

6. **Novelty by label.** A first visit to a prestated state or analyst-defined type is mistaken for creation of a new possibility.

7. **Self-confirming evidence.** Project papers and pipelines share assumptions and error modes; external replication is required.

8.3 Graded verdict

Table 2: Graded verdict. Ratings refer to ICP as a research principle, not to the truth of any ontology.

Dimension	Rating	Evidential basis and ceiling
Conceptual coherence	Strong	Formation, intervention, future family, and succession are separable and operationally specifiable.
Operational precision	Moderate–strong	The carrier gates and matched contrasts can fail; the signed ledger handles pruning and enabling; no accepted invariant yet exists.
Distinctness from prior art	Partial	Succession beyond the original carrier is narrower than reachability or adjacent-possible expansion, but complete-state, causal-state, and semantic-information accounts remain live.
Biological applicability	Moderate	The criterion classifies a known chromatin memory correctly and identifies an untested role-succession experiment.
Project evidence	Bounded	Artificial-life interventions give constructive examples, not observations of living systems.
Falsifiability	Strong locally	Carrier edits, state augmentation, representation changes, and held-out interventions can reject individual claims.

ICP is a coherent, falsifiable operational hypothesis about persistent organization. Existing science supplies strong precedents for each of its components but not for their integration. It is a viable research principle, not an established law.

9 Research path and conclusion

The program should advance in an order in which each step can fail.

Role succession in a living system. Carry out the prospective *INO1* test of Section 6, and its analogues in other systems with experimentally separable record and role: prion conformational inheritance, CRISPR spacer acquisition and loss, and enzyme networks with measured turnover. Freeze the role equivalence, horizon, and shams before opening outcomes, and fit the best ordinary model on training interventions before evaluating both on held-out ones.

Replication across artificial laws. Repeat the testbed's succession protocols under alternative local laws, schedulers, topologies, and independent implementations, requiring Eq. (3) to keep its qualitative result.

Adjacent-frontier assay. On each system, freeze the type map, actualized repertoire, policies, and horizons; report enabled, lost, and retained identities (Eq. (6)) rather than novelty counts; and compare against fixed-state, reinforced-urn, branching, and TAP rivals. Test the carrier dependence of the frontier (Section 4) by removing and restoring a carrier after its novelty event, and attempt to predict a system's triggering rate from carrier measurements before fitting its sequence. The strongest near-term signature is a prospectively predicted crossover of the form of Eq. (8), with $h_\star$ forecast from separately measured carrier lifetimes.

Stopping rules. The principle should be narrowed or reformulated if complete-state augmentation repeatedly removes the candidate historical variable, if carrier interventions fail under matched controls, if the role disappears under reasonable representation changes, or if no compression, invariant, prohibition, or prediction emerges beyond native theories. A result that reduces ICP to a precise theory of physically embodied memory would still be valuable.

Conclusion. Information can persist beyond the object that first carried it, but every established example does so through continuing physical realization in another state, relation, or organization. Biology shows records outliving their initiators, constraints maintained through the turnover of their realizers, and futures opened by prior organization. ICP's contribution is a single operational test, with a signed formalism for pruning and enabling, that makes these cases comparable and that separates the persistence of a record from the persistence of a role. Its claim to be more than an

organizing vocabulary rests on a result it has not yet produced: a held-out prediction, from carrier succession, that the best complete-state model of the same system misses.

Data and code availability

The literature registry, exact search log, claim–evidence matrix, rival and falsification registers, formal derivations, and manuscript sources are available at https://github.com/fmertz55/ICP. The simulation code, preregistrations, and frozen outputs of the companion artificial-life study are available at https://github.com/fmertz55/iota.

Ethics statement

This theoretical and computational study used no human participants, human tissue, animals, or field samples; ethics approval and consent were not applicable.

Funding

This research received no external funding.

Declaration of competing interests

The author declares no known competing financial interests or personal relationships that could have appeared to influence the work reported here.

CRediT author statement

Douglas Clark: Conceptualization; Methodology; Formal analysis; Investigation; Data curation; Software; Validation; Visualization; Writing – original draft; Writing – review and editing; Project administration.

Declaration of generative AI and AI-assisted technologies

During preparation of this work, the author used OpenAI Codex and Anthropic Claude for AI-assisted literature triage, drafting, code review, and LaTeX editing. The author reviewed and verified the cited sources, calculations, code, claims, and final text and takes full responsibility for the content of the publication.

References

[1] Nihat Ay and Daniel Polani. Information flows in causal networks. *Advances in Complex Systems*, 11:17–41, 2008. doi: 10.1142/S0219525908001465.

[2] Nix Barnett and James P. Crutchfield. Computational mechanics of input–output processes: Structured transformations and the epsilon-transducer. *Journal of Statistical Physics*, 161: 404–451, 2015. doi: 10.1007/s10955-015-1327-5. URL https://arxiv.org/abs/1412.2690.

[3] Rodolphe Barrangou et al. CRISPR provides acquired resistance against viruses in prokaryotes. *Science*, 315:1709–1712, 2007. doi: 10.1126/science.1138140.

[4] Alessandro Bellina, Giordano De Marzo, and Vittorio Loreto. Full spectrum of novelties in time-dependent urn models. *Physical Review Research*, 7:023127, 2025. doi: 10.1103/PhysRevResearch.7.023127.

[5] Douglas Clark. Constraint transport and history-dependent successor accessibility in IOTA: An information-first finite local physics model. Manuscript in preparation for submission; project evidence, 2026. URL https://github.com/fmertz55/iota.

[6] Luca Corti. Organizational normativity and teleology: A critique. *Synthese*, 202:96, 2023. doi: 10.1007/s11229-023-04255-7.

[7] David Deutsch. Constructor theory. *Synthese*, 190:4331–4359, 2013. doi: 10.1007/s11229-013-0279-z. URL https://arxiv.org/abs/1210.7439.

[8] David Deutsch and Chiara Marletto. Constructor theory of information. *Proceedings of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 471(2174):20140540, 2015. doi: 10.1098/rspa.2014.0540. URL https://arxiv.org/abs/1405.5563.

[9] Gabriele Di Bona, Alessandro Bellina, Giordano De Marzo, Angelo Petralia, Iacopo Iacopini, and Vito Latora. The dynamics of higher-order novelties. *Nature Communications*, 16:393, 2025. doi: 10.1038/s41467-024-55115-y.

[10] Ian B. Dodd, Mille A. Micheelsen, Kim Sneppen, and Geneviève Thon. Theoretical analysis of epigenetic cell memory by nucleosome modification. *Cell*, 129(4):813–822, 2007. doi: 10.1016/j.cell.2007.02.053.

[11] Richard A. Fariña and Sebastián Vallejo. The biological cosmological constant Λ_B: Exploratory propensity, dynamical habitability and the geometric origin of life. *BioSystems*, 265:105816, 2026. doi: 10.1016/j.biosystems.2026.105816.

[12] Keith D. Farnsworth. How an information perspective helps overcome the challenge of biology to physics. *BioSystems*, 217:104683, 2022. doi: 10.1016/j.biosystems.2022.104683.

[13] Justin Garson. Against organizational functions. *Philosophy of Science*, 84, 2017. doi: 10.1086/694009.

[14] James Griesemer. Modeling scaffolded development into the adjacent possible environment. *Philosophy of Science*, 91(5):1404–1414, 2024. doi: 10.1017/psa.2023.144.

[15] Michael J. Harms and Joseph W. Thornton. Historical contingency and its biophysical basis in glucocorticoid receptor evolution. *Nature*, 512:203–207, 2014. doi: 10.1038/nature13410.

[16] Iacopo Iacopini, Staša Milojević, and Vito Latora. Network dynamics of innovation processes. *Physical Review Letters*, 120(4):048301, 2018. doi: 10.1103/PhysRevLett.120.048301.

[17] Ryan G. James, Nix Barnett, and James P. Crutchfield. Information flows? a critique of transfer entropies. *Physical Review Letters*, 116:238701, 2016. doi: 10.1103/PhysRevLett.116.238701.

[18] Stuart A. Kauffman. *Investigations*. Oxford University Press, New York, 2000. doi: 10.1093/oso/9780195121049.001.0001.

[19] Stuart A. Kauffman. Prolegomenon to patterns in evolution. *BioSystems*, 123:3–8, 2014. doi: 10.1016/j.biosystems.2014.03.004.

[20] Alexander S. Klyubin, Daniel Polani, and Chrystopher L. Nehaniv. Empowerment: A universal agent-centric measure of control. In *2005 IEEE Congress on Evolutionary Computation*, pages 128–135, 2005. doi: 10.1109/CEC.2005.1554676.

[21] Artemy Kolchinsky and David H. Wolpert. Semantic information, autonomous agency and non-equilibrium statistical physics. *Interface Focus*, 8:20180041, 2018. doi: 10.1098/rsfs.2018.0041.

[22] Solomon Kullback and Richard A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951. doi: 10.1214/aoms/1177729694.

[23] William H. Light et al. Interaction of a DNA zip code with the nuclear pore complex promotes H2A.Z incorporation and INO1 transcriptional memory. *Molecular Cell*, 40, 2010. doi: 10.1016/j.molcel.2010.09.007.

[24] Dibyendu Mandal and Christopher Jarzynski. Work and information processing in a solvable model of Maxwell's demon. *Proceedings of the National Academy of Sciences*, 109:11641–11645, 2012. doi: 10.1073/pnas.1204263109.

[25] Ian M. Mitchell, Alexandre M. Bayen, and Claire J. Tomlin. A time-dependent Hamilton–Jacobi formulation of reachable sets for continuous dynamic games. *IEEE Transactions on Automatic Control*, 50(7):947–957, 2005. doi: 10.1109/TAC.2005.851439.

[26] Bernardo Monechi, Álvaro Ruiz-Serrano, Francesca Tria, and Vittorio Loreto. Waves of novelties in the expansion into the adjacent possible. *PLOS ONE*, 12(6):e0179303, 2017. doi: 10.1371/journal.pone.0179303.

[27] Ma"el Montévil and Matteo Mossio. Biological organisation as closure of constraints. *Journal of Theoretical Biology*, 372:179–191, 2015. doi: 10.1016/j.jtbi.2015.02.029.

[28] Sudip Patra and Stuart A. Kauffman. Domains of laws yet domains of no law: Energy and work, responsible free will choice, and doing. *BioSystems*, 253:105501, 2025. doi: 10.1016/j.biosystems.2025.105501.

[29] Takahiro Sagawa and Masahito Ueda. Generalized Jarzynski equality under nonequilibrium feedback control. *Physical Review Letters*, 104:090602, 2010. doi: 10.1103/PhysRevLett.104.090602.

[30] Patrick Saint-Pierre. Approximation of the viability kernel. *Applied Mathematics and Optimization*, 29:187–209, 1994. doi: 10.1007/BF01204182.

[31] Cosma Rohilla Shalizi and James P. Crutchfield. Computational mechanics: Pattern and prediction, structure and simplicity. *Journal of Statistical Physics*, 104:817–879, 2001. doi: 10.1023/A:1010388907793. URL https://arxiv.org/abs/cond-mat/9907176.

[32] Mike Steel, Wim Hordijk, and Stuart A. Kauffman. Dynamics of a birth–death process based on combinatorial innovation. *Journal of Theoretical Biology*, 491:110187, 2020. doi: 10.1016/j.jtbi.2020.110187.

[33] Brandon Sump et al. Mitotically heritable, RNA polymerase II-independent H3K4 dimethylation stimulates INO1 transcriptional memory. *eLife*, 11:e77646, 2022. doi: 10.7554/eLife.77646.

[34] Josef Taalbi. Long-run patterns in the discovery of the adjacent possible. *Industrial and Corporate Change*, 35(1):123–149, 2026. doi: 10.1093/icc/dtaf028.

[35] Shoichi Toyabe, Takahiro Sagawa, Masahito Ueda, Eiro Muneyuki, and Masaki Sano. Experimental demonstration of information-to-energy conversion and validation of the generalized Jarzynski equality. *Nature Physics*, 6:988–992, 2010. doi: 10.1038/nphys1821.

[36] Francesca Tria, Vittorio Loreto, Vito D. P. Servedio, and Steven H. Strogatz. The dynamics of correlated novelties. *Scientific Reports*, 4:5890, 2014. doi: 10.1038/srep05890.

[37] Enrico Ubaldi, Raffaella Burioni, Vittorio Loreto, and Francesca Tria. Emergence and evolution of social networks through exploration of the adjacent possible space. *Communications Physics*, 4:28, 2021. doi: 10.1038/s42005-021-00527-1.

[38] Nilesh Vaidya et al. Spontaneous network formation among cooperative RNA replicators. *Nature*, 491:72–77, 2012. doi: 10.1038/nature11549.

[39] Vera Vasas, E"ors Szathmáry, and Mauro Santos. Lack of evolvability in self-sustaining autocatalytic networks constraints metabolism-first scenarios for the origin of life. *Proceedings of the National Academy of Sciences*, 107, 2010. doi: 10.1073/pnas.0912628107.

[40] Daniel M. Weinreich, Nigel F. Delaney, Mark A. DePristo, and Daniel L. Hartl. Darwinian evolution can follow only very few mutational paths to fitter proteins. *Science*, 312:111–114, 2006. doi: 10.1126/science.1123539.

[41] Alexander D. Wissner-Gross and Cameron E. Freer. Causal entropic forces. *Physical Review Letters*, 110:168702, 2013. doi: 10.1103/PhysRevLett.110.168702.